

YEAR 7 MATHEMATICS

Chapter 1

Integers

MIXED PRACTICE WORKSHEETS

1

How to use this booklet

- Each lesson is **one double-sided sheet**. Write your name, class and date at the top.
- Then work down the **columns**, left to right. They get harder as you go: **Easy** **Easy** **Medium** on page 1, then **Medium** **Challenging** **Challenging** on page 2.
- Most questions practise **today's skill**. A question marked **from 1.03** comes from an earlier lesson, to keep it fresh. Stuck? Look back at that lesson.
- Each question tells you what to do: read it carefully. Show your working on the lines. The first question with working in each Easy column shows you how to set it out.

WE DO

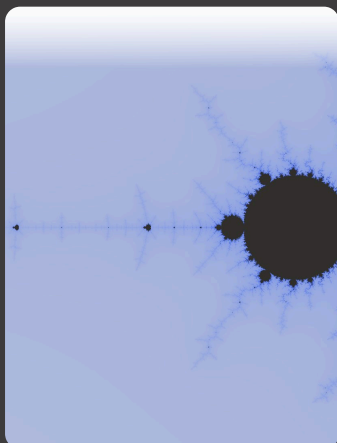
The grey box at the top. Your teacher shows the key idea with two or three examples. **Copy every step** into the space.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
1.01	Integers and opposites	3		☐
1.02	The number line	5		☐
1.03	Comparing and ordering integers	7		☐
1.04	Adding integers	9		☐
1.05	Subtracting integers	11		☐
1.06	Multiplying integers	13		☐
1.07	Dividing integers	15		☐
1.08	Order of operations with integers	17		☐
1.09	Integers on a calculator	19		☐
1.10	Integer problems	21		☐

In this chapter you will

- understand positive and negative numbers and their opposites
- use a number line to show and compare integers
- order integers
- add, subtract, multiply and divide integers
- use the order of operations with integers, with and without a calculator
- solve problems involving integers



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the long antenna of the set along the negative real axis, dotted with tiny copies of the whole set. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

An integer is a whole number: positive, negative or zero. The opposite of a number is the same distance from 0, on the other side of 0.

Watch your teacher.
Copy every step.

E1 Write an integer for 5 m below sea level.

E2 What is the opposite of -6?

E3 $-(-7)$

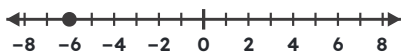
Easy

Questions 1-9

1 Write an integer for spending \$36.

2 What is the opposite of -20?

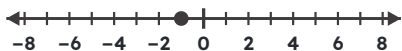
3 Which integer is at the dot?



4 Write an integer for winning 10 points.

5 What is the opposite of 27?

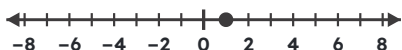
6 Which integer is at the dot?



7 Write an integer for spending \$18.

8 What is the opposite of 25?

9 Which integer is at the dot?

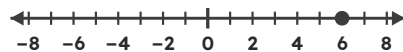


Easy

Questions 10-18

10 What is the opposite of 4?

11 Which integer is at the dot?



12 Write an integer for a drop of 10 cm.

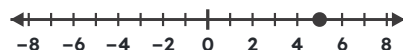
13 $-(-12)$

= the opposite of ____

= ____

14 What is the opposite of 19?

15 Which integer is at the dot?



16 Write an integer for a deposit of \$34.

17 $-(-1)$

18 What is the opposite of -27?

Medium

Questions 19-26

19 $-(-17)$

20 Which is further from 0: -14 or 6?

21 Which two integers are 37 steps from 0?

22 The opposite of "8 up" is "8 down". Write both as integers.

23 $-(+15)$

24 Which is further from 0: -3 or 10?

25 Which two integers are 7 steps from 0?

26 The opposite of "12 forwards" is "12 backwards". Write both as integers.

Medium

Questions 27–37

- 27 Which is further from 0: -9 or 10 ?

- 28 Which two integers are 40 steps from 0?

- 29 The opposite of "11 forwards" is "11 backwards". Write both as integers.

- 30 $-(-18)$

- 31 Two opposites are 28 steps apart. What are they?

- 32 Which is further from 0: -9 or 2 ?

- 33 Which two integers are 13 steps from 0?

- 34 $5 \times 6 - 5 \times 5$

from Year 6

- 35 The opposite of "16 up" is "16 down". Write both as integers.

- 36 $-(+16)$

- 37 Two opposites are 44 steps apart. What are they?

Challenging

Questions 38–47

- 38 Two opposites are 34 steps apart. What are they?

- 39 $-(-(-13))$

- 40 Mia says, "The opposite of a number is always negative." Is Mia right? Explain.

- 41 Two opposites are 58 steps apart. What are they?

- 42 $-(-(-15))$

- 43 What is the opposite of 0? Explain.

- 44 Two opposites are 52 steps apart. What are they?

- 45 $100 - 39$

from Year 6

- 46 $-(-(-14))$

- 47 Ben says $-(-4) = -4$. Is Ben right? Explain.

Challenging

Questions 48–57

- 48 $-(-(-10))$

- 49 Which integers from -6 to 6 are more than 4 steps from 0?

- 50 Two opposites are 38 steps apart. What are they?

- 51 The opposite of "15 forwards" is "15 backwards". Write both as integers.

- 52 $-(-(-11))$

- 53 A number is 3 steps from 0 and less than 0. What is its opposite?

- 54 Two opposites are 36 steps apart. What are they?

- 55 Which is largest: 7 671, 6 680, 5 782, 1 812?

from Year 6

- 56 The opposite of "16 forwards" is "16 backwards". Write both as integers.

- 57 $-(-(-8))$

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Numbers get larger to the right. To find a distance, count the gaps. If the trip passes 0, count to 0, then count on.

Watch your teacher.
Copy every step.

E1 How far is it from -3 to 5 ?

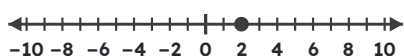
E2 Start at 4 . Move 9 left. Where do you land?

E3 Which integer is halfway between -8 and 2 ?

Easy

Questions 1–9

1 Which integer is at the dot?

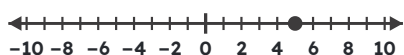


2 Start at 0 . Move 2 left. Where do you land?

3 How far is it from -5 to 7 ?

4 Write the integers between -6 and -2 .

5 Which integer is at the dot?

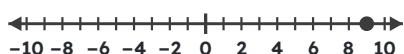


6 Start at 0 . Move 7 right. Where do you land?

7 How far is it from -8 to 4 ?

8 Write the integers between -5 and -1 .

9 Which integer is at the dot?



Easy

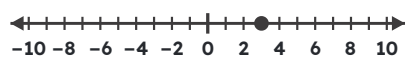
Questions 10–18

10 Start at 4 . Move 9 right. Where do you land?

11 How far is it from -3 to 9 ?

12 Write the integers between -7 and -2 .

13 Which integer is at the dot?



14 How far is it from -13 to 16 ?

$$\underline{\quad} + \underline{\quad} = \underline{\quad}$$

15 What is the opposite of -19 ?

from 1.01

16 Start at -6 . Move 7 left. Where do you land?

17 How far is it from -8 to 9 ?

18 Write the integers between -6 and -3 .

Medium

Questions 19–26

19 How far is it from -9 to 16 ?

20 Which integer is halfway between -10 and 6 ?

21 A number line counts by 5 s: -10 , -5 , 0 , ... What are the next two numbers?

22 A lift goes from level -2 to level 7 . How many floors does it go up?

23 It is 10°C . It gets 14 degrees colder. What is the temperature now?

from Year 6

24 Point A is at -5 . Point B is 14 to the left of A. Where is B?

25 How far is it from -11 to 19 ?

26 Which integer is halfway between -15 and 1 ?

Medium

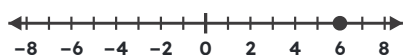
Questions 27–37

- 27 Which integer is halfway between -11 and 5 ?

- 28 A number line counts by 5s: $-15, -10, -5, \dots$ What are the next two numbers?

- 29 A lift goes from level -3 to level 2 . How many floors does it go up?

- 30 Which integer is at the dot?



from 1.01

- 31 Point A is at 1 . Point B is 7 to the left of A. Where is B?

- 32 How far is it from -24 to 19 ?

- 33 Two integers are 12 apart. Halfway between them is -9 . What are they?

- 34 The opposite of "15 above zero" is "15 below zero". Write both as integers.

from 1.01

- 35 Which integer is halfway between -12 and -6 ?

- 36 A number line counts by 10s: $-20, -10, 0, \dots$ What are the next two numbers?

- 37 What is the value of the 2 in 47 269?

from Year 6

Challenging

Questions 38–46

- 38 Two integers are 12 apart. Halfway between them is -3 . What are they?

- 39 Start at 2 . Move 6 right, then 11 left. Where do you land?

- 40 How many integers are between -12 and 3 ? Do not count -12 and 3 .

- 41 What is the opposite of 16 ?

from 1.01

- 42 A diver is at -15 m. A fish is at -6 m. How far above the diver is the fish?

- 43 Two integers are 14 apart. Halfway between them is -7 . What are they?

- 44 Start at 1 . Move 5 right, then 15 left. Where do you land?

- 45 Which is further from 0: -5 or 14 ?

from 1.01

- 46 How many integers are between -4 and 7 ? Do not count -4 and 7 .

Challenging

Questions 47–56

- 47 Start at 5 . Move 7 right, then 9 left. Where do you land?

- 48 How many integers are between -6 and 8 ? Do not count -6 and 8 .

- 49 A diver is at -20 m. A fish is at 1 m. How far above the diver is the fish?

- 50 What is $\frac{4}{8}$ of 80 ?

from Year 6

- 51 Two integers are 6 apart. Halfway between them is -3 . What are they?

- 52 Point A is at 5 . Point B is 9 to the left of A. Where is B?

- 53 Start at -2 . Move 5 right, then 15 left. Where do you land?

- 54 Write an integer for a loss of 18 kg.

from 1.01

- 55 How many integers are between -5 and 6 ? Do not count -5 and 6 .

- 56 A diver is at -10 m. A fish is at -1 m. How far above the diver is the fish?

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

The number further right on the number line is larger. So $-2 > -7$, because -2 is further right. Ascending: smallest to largest.

Watch your teacher.
Copy every step.

E1 Write $<$ or $>$: -9 ☐ -4

E2 Order from smallest to largest: 3, -5 , 0, -1

E3 Which is colder: -12°C or -8°C ?

Easy

Questions 1–9

1 Write $<$ or $>$: -10 ☐ -6

2 Which is larger: 3 or -5 ?

3 Which is smaller: -8 or -18 ?

4 Which is colder: -8°C or -10°C ?

5 Write $<$ or $>$: -10 ☐ -7

6 Which is larger: 1 or -18 ?

7 Which is smaller: 2 or -6 ?

8 Which is colder: -7°C or -10°C ?

9 Start at 6. Move 5 left. Where do you land?

from 1.02

Easy

Questions 10–18

10 Which is larger: -9 or -15 ?

11 Which is smaller: -15 or -2 ?

12 Which is colder: -2°C or -15°C ?

13 $100 - 29$

from Year 6

14 Write $<$ or $>$: 4 ☐ -10

15 Order from smallest to largest: -6 , 12, 3, 0, -8

smallest: ___

16 Which is larger: -10 or 1?

17 What is the opposite of 1?

from 1.01

18 Which is smaller: -19 or 1?

Medium

Questions 19–26

19 Order from smallest to largest: -4 , 5, 0, -11 , 10

20 Order from largest to smallest: 6, -6 , -18 , -28

21 A number line counts by 10s: -30 , -20 , -10 , ... What are the next two numbers?

from 1.02

22 Write $<$ or $>$: -13 ☐ -44

23 Write all the integers between -9 and -6 .

24 $42 \div 6$

from Year 6

25 Order from smallest to largest: 0, -3 , 6, -12 , 8

26 Order from largest to smallest: -23 , -15 , 13, -4

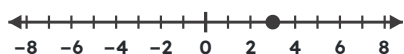
Medium

Questions 27–37

- 27 Order from largest to smallest:
-27, -10, -2, 9

- 28 Write $<$ or $>$: $-47 \square -20$

- 29 Which integer is at the dot?



from 1.01

- 30 Write all the integers between -8 and -5.

- 31 A lift goes from level -2 to level 9. How many floors does it go up?

from 1.02

- 32 Order from smallest to largest:
-11, 9, -3, 6, 0

- 33 Dawn temperatures: Oberon 8°C , Cooma -10°C , Jindabyne 0°C , Thredbo -3°C . Which town was coldest? Which was warmest?

- 34 What is $\frac{5}{8}$ of 80?

from Year 6

- 35 Order from largest to smallest:
-27, -15, 5, -7

- 36 Write an integer for 38 m above sea level.

from 1.01

- 37 Write $<$ or $>$: $-19 \square -31$

Challenging

Questions 38–47

- 38 Dawn temperatures: Oberon -3°C , Cooma 5°C , Jindabyne -7°C , Thredbo 0°C . Which town was coldest? Which was warmest?

- 39 True or false: $-9 > -12$. Explain.

- 40 Point A is at -10. Point B is 8 to the left of A. Where is B?

from 1.02

- 41 Write the smallest and the largest integer between -15 and 5.

- 42 $38 - 10 \div 5$

from Year 6

- 43 Sam says -8 is larger than -3 because 8 is larger than 3. Is Sam right? Explain.

- 44 Dawn temperatures: Oberon 5°C , Cooma -1°C , Jindabyne 0°C , Thredbo -10°C . Which town was coldest? Which was warmest?

- 45 What is the opposite of -6?

from 1.01

- 46 True or false: $-6 > -10$. Explain.

- 47 How far is it from -18 to 37?

from 1.02

Challenging

Questions 48–57

- 48 True or false: $-8 < -11$. Explain.

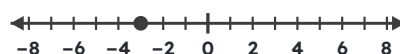
- 49 Write the smallest and the largest integer between -15 and 2.

- 50 Which is largest: 9 269, 6 177, 7 046, 6 408?

from Year 6

- 51 Order from coldest to warmest: 2°C , -11°C , -4°C , 0°C , -15°C

- 52 Which integer is at the dot?



from 1.01

- 53 Dawn temperatures: Oberon -6°C , Cooma -3°C , Jindabyne 0°C , Thredbo 4°C . Which town was coldest? Which was warmest?

- 54 Write all the integers between -4 and 0.

- 55 Which integer is halfway between -13 and 1?

from 1.02

- 56 True or false: $-2 > -10$. Explain.

- 57 $600 + 193$

from Year 6

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Adding a positive moves right. Adding a negative moves left: $+$ (-4) means $- 4$. A $+$ and a $-$: take the difference, and the sign with more wins.

Watch your teacher.
Copy every step.

E1 $-3 + 5$

E2 $4 + (-9)$

E3 $-6 + (-2)$

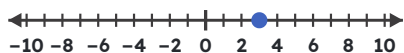
Easy

Questions 1–12

1 $-2 + 9$

2 $3 + (-9)$

3 Use the number line to find $3 + 5$.

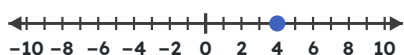


4 $-7 + (-8)$

5 $-1 + 6$

6 $9 + (-6)$

7 Use the number line to find $4 + 3$.



8 $-9 + (-8)$

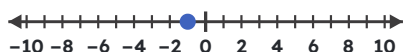
9 Start at 3. Move 4 left. Where do you land?

from 1.02

10 $-9 + 9$

11 $8 + (-2)$

12 Use the number line to find $-1 + 3$.

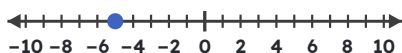


Easy

Questions 13–21

13 $4 + (-7)$

14 Use the number line to find $-5 + (-2)$.



15 $-7 + (-4)$

16 $70 \div 7$

from Year 6

17 $-8 + 7$

18 $-38 + (-19)$

$= ___ - ___$

$= ___$

19 $1 + (-8)$

20 Which is larger: -17 or -3 ?

from 1.03

21 Use the number line to find $0 + 1$.



Medium

Questions 22–29

22 $-19 + (-17)$

23 $15 + (-44)$

24 Which is further from 0: -2 or 13 ?

from 1.01

25 It is -12°C . The temperature rises 13 degrees. What is it now?

26 $-6 + 3 + (-10)$

27 What is the value of the 3 in 30 458?

from Year 6

28 $-27 + (-18)$

29 $19 + (-43)$

Medium

Questions 30–40

30 $9 + (-32)$

31 It is -10°C . The temperature rises 8 degrees. What is it now?

32 How far is it from -5 to 9 ?

from 1.02

33 $-7 + 7 + (-11)$

34 Write $<$ or $>$: $-21 \square -20$

from 1.03

35 $-27 + (-24)$

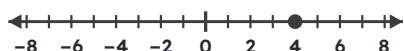
36 Find the missing number:
 $-5 + \square = -8$

37 $7 \times 7 - 2 \times 4$

from Year 6

38 $27 + (-38)$

39 Which integer is at the dot?



from 1.01

40 It is -9°C . The temperature rises 13 degrees. What is it now?

Challenging

Questions 41–50

41 Find the missing number:
 $-10 + \square = -6$

42 Kai has \$47, spends \$90, then earns \$27. What is the balance now?

43 A lift goes from level -1 to level 8. How many floors does it go up?

from 1.02

44 Ruby says $-3 + (-7) = 10$. Is Ruby right? Explain.

45 Write 0.1 as a fraction.

from Year 6

46 A submarine at -100 m rises 100 m, then sinks 70 m. How deep is it now?

47 $-11 + 5 + (-16) + 2$

48 Which is colder: -6°C or -11°C ?

from 1.03

49 Find the missing number:
 $-12 + \square = -5$

50 The opposite of "19 up" is "19 down". Write both as integers.

from 1.01

Challenging

Questions 51–60

51 Leo has \$54, spends \$65, then earns \$22. What is the balance now?

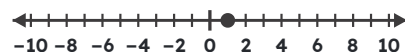
52 Tom says $-8 + (-6) = 14$. Is Tom right? Explain.

53 It is 10°C . It gets 15 degrees colder. What is the temperature now?

from Year 6

54 A lift is at level -1 . It goes up 5 floors, then down 3. Which level is it on now?

55 Which integer is at the dot?



from 1.02

56 $-9 + 12 + (-13) + 4$

57 Find the missing number:
 $-4 + \square = -2$

58 Order from smallest to largest: 0, -2 , -9 , 1, 8

from 1.03

59 $4 + 2 + (-10)$

60 10×11

from Year 6

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Subtracting a positive moves left. Subtracting a negative moves right: $-(-3)$ means

Watch your teacher. Copy every step.

E1 $2 - 7$

E2 $5 - (-3)$

E3 $-4 - (-9)$

Easy

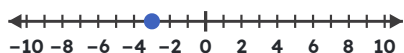
Questions 1-12

1 $1 - 7$

2 $1 - (-8)$

3 $-5 - 6$

4 Use the number line to find $-3 - (-1)$.

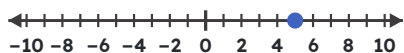


5 $2 - 6$

6 $4 - (-7)$

7 $-7 - 5$

8 Use the number line to find $5 - (-2)$.



9 $8 + (-3)$

from 1.04

10 $4 - 8$

11 $3 - (-2)$

12 $-6 - 5$

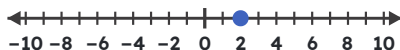
Easy

Questions 13-22

13 $3 - (-5)$

14 $-9 - 7$

15 Use the number line to find $2 - 5$.



16 Write 0.1 as a fraction.

from Year 6

17 $4 - 6$

18 $-17 - (-7)$

= ____ + ____

= ____

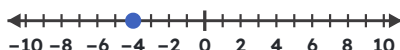
19 $3 - (-8)$

20 Which is larger: 3 or -5?

from 1.03

21 $-3 - 3$

22 Use the number line to find $-4 - (-2)$.



Medium

Questions 23-29

23 $-11 - (-21)$

24 The temperature went from -10°C to 13°C . How much did it rise?

25 Which is further from 0: -4 or 10?

from 1.01

26 Find the difference between 28 and -26.

27 $-37 - (-49)$

28 Which integer is halfway between -8 and -2?

from 1.02

29 $-14 - (-6)$

Medium

Questions 30–39

- 30 The temperature went from -8°C to 4°C . How much did it rise?

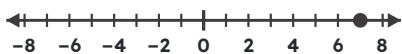
- 31 Find the difference between 12 and -10 .

- 32 $22 - 24 \div 4$

from Year 6

- 33 $-22 - (-6)$

- 34 Which integer is at the dot?

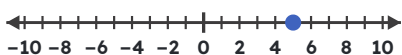


from 1.01

- 35 $-13 - (-6)$

- 36 Find the missing number:
 $\square - (-4) = 0$

- 37 Use the number line to find $5 + 5$.



from 1.04

- 38 The temperature went from -2°C to 15°C . How much did it rise?

- 39 Write $<$ or $>$: $-28 \square -11$

from 1.03

Challenging

Questions 40–49

- 40 Find the missing number:
 $\square - (-2) = 1$

- 41 $2 - (-5) - 5$

- 42 Which is largest: 1 898, 6 667, 7 624, 6 433?

from Year 6

- 43 A submarine is at -180 m. A plane is at 760 m. How far apart are they?

- 44 How far is it from -8 to 3 ?

from 1.02

- 45 Leo says $6 - (-4) = 2$. Is Leo right? Explain.

- 46 Find the missing number:
 $\square - (-3) = 5$

- 47 The opposite of "5 up" is "5 down". Write both as integers.

from 1.01

- 48 $-2 - (-6) - 10$

- 49 $-7 + 5 + (-7)$

from 1.04

Challenging

Questions 50–59

- 50 $-4 - (-4) - 5$

- 51 A submarine is at -190 m. A plane is at 320 m. How far apart are they?

- 52 $121 \div 11$

from Year 6

- 53 Aisha says $5 - (-7) = -2$. Is Aisha right? Explain.

- 54 What is the opposite of -15 ?

from 1.01

- 55 Find the missing number:
 $\square - (-8) = 9$

- 56 $-35 - (-23)$

- 57 Which is colder: -4°C or -19°C ?

from 1.03

- 58 $-2 - (-4) - 14$

- 59 A lift goes from level -1 to level 6 . How many floors does it go up?

from 1.02

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Multiply the numbers, then find the sign. Same signs: the answer is positive. Different signs: the answer is negative.

Watch your teacher.
Copy every step.

E1 -4×3

E2 $-5 \times (-6)$

E3 $-2 \times 3 \times (-4)$

Easy

Questions 1–12

1 -5×3

2 $-2 \times (-3)$

3 Positive or negative? Do not work it out: 7×11

4 Write $5 \times (-4)$ as a sum, then find the answer.

5 $3 \times (-4)$

6 $-9 \times (-9)$

7 Positive or negative? Do not work it out: 4×7

8 Write $3 \times (-3)$ as a sum, then find the answer.

9 $9 + (-7)$

from 1.04

10 -3×3

11 $-8 \times (-6)$

12 Positive or negative? Do not work it out: -8×12

Easy

Questions 13–21

13 $-8 \times (-9)$

14 Positive or negative? Do not work it out: 4×12

15 Write $2 \times (-2)$ as a sum, then find the answer.

$(\quad) + (\quad)$

16 $28 - 12 \div 4$

from Year 6

17 -8×8

18 $-15 \times (-3)$

19 $-5 \times (-8)$

20 $7 - (-5)$

from 1.05

21 Positive or negative? Do not work it out: 5×11

Medium

Questions 22–29

22 $-21 \times (-6)$

23 $1 \times (-1) \times 5$

24 Which is further from 0: -4 or 11?

from 1.01

25 $(-12)^2$

26 The temperature falls 3°C each hour for 5 hours. Write a multiplication for the change.

27 Order from largest to smallest: 3, -21 , -10 , -3

from 1.03

28 $-18 \times (-4)$

29 $-4 \times (-3) \times (-5)$

Medium

Questions 30–40

30 $2 \times (-4) \times (-3)$

31 $(-3)^2$

32 Write 0.9 as a fraction.

from Year 6

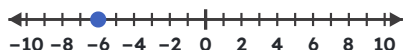
33 The temperature falls 5°C each hour for 3 hours. Write a multiplication for the change.

34 Start at -3 . Move 5 right. Where do you land?

from 1.02

35 $-21 \times (-4)$

36 Find the missing number:
 $\square \times (-7) = -49$

37 Use the number line to find $-6 + 1$.


from 1.04

38 $-2 \times 1 \times 2$

39 Find the difference between 26 and -10 .

from 1.05

40 $(-2)^2$

Challenging

Questions 41–50

41 Find the missing number:
 $\square \times 2 = 16$

42 Find two integers that multiply to -56 and add to -1 .

43 It is 8°C . It gets 16 degrees colder. What is the temperature now?

from Year 6

44 $-1 \times (-1) \times (-1) \times (-1) \times (-1) \times (-1)$

45 Which is smaller: -12 or -2 ?

from 1.03

46 Ruby says $(-8)^2 = -64$. Is Ruby right? Explain.

47 Find the missing number:
 $\square \times (-5) = 30$

48 Which two integers are 19 steps from 0?

from 1.01

49 Find two integers that multiply to -48 and add to 2.

50 $-6 + 3 + (-8)$

from 1.04

Challenging

Questions 51–60

51 Find two integers that multiply to -40 and add to 3.

52 $-1 \times (-1) \times (-1) \times (-1) \times (-1) \times (-1) \times (-1)$

53 What is the value of the 1 in 83 175?

from Year 6

54 Tom says $(-8)^2 = -64$. Is Tom right? Explain.

55 How far is it from -2 to 5?

from 1.02

56 Find the missing number:
 $\square \times (-7) = 63$

57 The temperature falls 6°C each hour for 6 hours. Write a multiplication for the change.

58 Use the number line to find $-1 - 1$.


from 1.05

59 Find two integers that multiply to -8 and add to 2.

60 Write all the integers between -8 and -4 .

from 1.03

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Divide the numbers, then find the sign. The rules are the same as for $\times$: same signs give $+$, different signs give $-$.

Watch your teacher.
Copy every step.

E1 $-20 \div 4$

E2 $-36 \div (-9)$

E3 Find the average: $-3, 5, -8, -6$

Easy

Questions 1–12

1 $-24 \div 3$

2 $-18 \div (-2)$

3 $45 \div (-5)$

4 Positive or negative? Do not work it out: $6 \div (-6)$

5 $-36 \div 4$

6 $-21 \div (-7)$

7 $12 \div (-6)$

8 Positive or negative? Do not work it out: $-24 \div 3$

9 $-7 \times (-7)$

from 1.06

10 $-8 \div 2$

11 $-18 \div (-9)$

12 $28 \div (-7)$

Easy

Questions 13–22

13 $-28 \div (-4)$

14 $49 \div (-7)$

15 Positive or negative? Do not work it out: $-24 \div (-8)$

16 Write 0.3 as a fraction.

from Year 6

17 $-49 \div 7$

18 $-77 \div (-7)$

same signs: _____

= ____

19 $-8 \div (-2)$

20 $-1 - (-4)$

from 1.05

21 $18 \div (-9)$

22 Positive or negative? Do not work it out: $-12 \div (-3)$

Medium

Questions 23–29

23 $-72 \div 6$

24 Find the average of these temperatures: $-3^\circ\text{C}, 5^\circ\text{C}, 4^\circ\text{C}, -5^\circ\text{C}, -6^\circ\text{C}$

25 Which is further from 0: -11 or 14 ?

from 1.01

26 A diver goes down 42 m in 7 equal stages. Write a division for each stage.

27 Work out $\frac{-12}{-6}$

28 $22 + (-15)$

from 1.04

29 $-88 \div (-8)$

Medium

Questions 30–39

- 30 Find the average of these temperatures: -1°C , 6°C , -8°C , -1°C

- 31 A diver goes down 35 m in 7 equal stages. Write a division for each stage.

32 $22 - 20 \div 5$

from Year 6

33 Work out $\frac{-81}{9}$

- 34 Start at 1. Move 5 left. Where do you land?

from 1.02

35 $-55 \div (-5)$

- 36 Find the missing number:
 $\square \div 7 = 1$

- 37 Positive or negative? Do not work it out: -11×5

from 1.06

- 38 Find the average of these temperatures: -11°C , 3°C , 3°C , 5°C

- 39 Find the difference between 21 and -12 .

from 1.05

Challenging

Questions 40–49

- 40 Find the missing number:
 $\square \div 3 = -12$

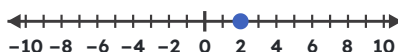
- 41 Find the missing number:
 $-12 \div \square = 4$

- 42 It is 9°C . It gets 14 degrees colder. What is the temperature now?

from Year 6

43 $-30 \div 5 \times (-2)$

- 44 Use the number line to find $2 + (-3)$.



from 1.04

- 45 Lena says $-49 \div (-7) = -7$. Is Lena right? Explain.

- 46 Find the missing number:
 $\square \div 3 = 5$

- 47 Order from largest to smallest: -23 , 3 , -13 , -7

from 1.03

- 48 Find the missing number:
 $-14 \div \square = 7$

- 49 The temperature falls 5°C each hour for 4 hours. Write a multiplication for the change.

from 1.06

Challenging

Questions 50–59

- 50 Find the missing number:
 $18 \div \square = -9$

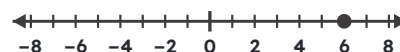
51 $4 \div (-1) \times (-4)$

52 $88 \div 11$

from Year 6

- 53 Leo says $-18 \div (-3) = -6$. Is Leo right? Explain.

- 54 Which integer is at the dot?

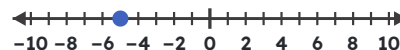


from 1.01

- 55 Find the missing number:
 $\square \div (-1) = 8$

56 Work out $\frac{-81}{-9}$

- 57 Use the number line to find $-5 - (-5)$.



from 1.05

- 58 Find the missing number:
 $-6 \div \square = 2$

59 $1 + 9 + (-7)$

from 1.04

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Brackets first. Then $\times$ and $\div$, from left to right. Then $+$ and $-$, from left to right. Write one step a line.

Watch your teacher.
Copy every step.

E1 $-3 + 4 \times (-2)$

E2 $(-6 + 2) \times 5$

E3 $-20 \div (3 - 8)$

Easy

Questions 1–9

1 $-4 + 3 \times (-1)$

$= ___ + (___)$

$= ___$

2 $4 \times (-3) - 5$

3 $(4 + 1) \times (-4)$

4 $9 + 3 \times 5$

5 $5 \times 6 - 10$

6 $(9 + (-7)) \times 4$

7 $7 + 4 \times 2$

8 $6 \times (-3) - 10$

9 $-9 \times (-4)$

from 1.06

Easy

Questions 10–18

10 $5 \times 2 - 12$

$= ___ - ___$

$= ___$

11 $(-8 + (-1)) \times 2$

12 $8 + 6 \times 4$

13 $24 - 18 \div 6$

from Year 6

14 $-8 \div 2 + (-5) \times 5$

15 $5 \times (-3) - 5$

16 $(-1 + 4) \times 1$

17 $-48 \div (-6)$

from 1.07

18 $1 + 3 \times 1$

Medium

Questions 19–26

19 $18 \div 3 + (-4) \times 3$

20 $-7 - (6 - (-7))$

21 Which is further from 0: -15 or 5?

from 1.01

22 $(-6)^2 - 30$

23 $5 \div (1 - 6)$

24 The temperature went from -5°C to 9°C . How much did it rise?

from 1.05

25 $-20 \div (-5) + 1 \times 5$

26 $-11 - (9 - (-2))$

Medium

Questions 27–37

27 $-8 - (4 - (-8))$

28 $(-3)^2 - 13$

29 What is the value of the 9 in 96 406?

from Year 6

30 $-6 \div (3 - 6)$

 31 Start at -2 . Move 3 left. Where do you land?

from 1.02

32 $2 \div 2 + (-5) \times 2$

33 $(-8 - (-9)) \times (1 + 6)$

 34 Positive or negative? Do not work it out: -2×12

from 1.06

35 $2 - (1 - 3)$

36 A diver goes down 96 m in 8 equal stages. Write a division for each stage.

from 1.07

37 $(-4)^2 - 19$

Challenging

Questions 38–47

38 $(6 - (-5)) \times (-5 + (-2))$

39 $[-7 + 2 \times 2] \div 3$

40 100 students sit in 4 equal rows. How many are in each row?

from Year 6

 41 Put brackets in to make this true: $-5 + 3 \times 3 = -6$

42 $-4 - 5$

from 1.05

 43 Zac says $7 - 2 \times (-4) = -20$. Is Zac right? Explain.

44 $(2 - 6) \times (6 + 6)$

 45 Order from largest to smallest: $-26, -16, 1, -5$

from 1.03

46 $[-8 + 3 \times (-4)] \div 4$

 47 The temperature falls 5°C each hour for 8 hours. Write a multiplication for the change.

from 1.06

Challenging

Questions 48–57

48 $[-3 + 5 \times (-3)] \div 3$

 49 Put brackets in to make this true: $-9 + 5 \times 2 = -8$

50 Write 0.7 as a fraction.

from Year 6

 51 Leo says $7 - 3 \times (-5) = -20$. Is Leo right? Explain.

52 $6 + (-2)$

from 1.04

53 $(-6 - 9) \times (6 + (-1))$

54 $-15 \div (0 - 3)$

 55 Positive or negative? Do not work it out: $10 \div 2$

from 1.07

56 $[-5 + 5 \times (-3)] \div 5$

57 $-35 - (-14)$

from 1.05

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Use the $(-)$ key for a negative number and the $-$ key to subtract. Estimate first, then check that the calculator answer is close.

Watch your teacher.
Copy every step.

E1 Which keys do you press for -8×5 ?

E2 Use a calculator: $-348 + 129$

E3 Estimate, then use a calculator: -394×21

Easy

Questions 1–9

1 Use a calculator: $-521 + 189$

2 Use a calculator: -19×11

3 Which keys do you press for $-7 + 2$?

4 Use a calculator: $-957 \div (-29)$

5 Use a calculator: $-433 + 553$

6 Use a calculator: -40×22

7 Which keys do you press for -6×4 ?

8 Use a calculator: $-506 \div (-23)$

9 $2 \times 1 - 7$

from 1.08

Easy

Questions 10–18

10 Use a calculator: -30×27

11 Which keys do you press for $-5 - 8$?

12 Use a calculator: $-627 \div (-19)$

13 $88 \div 8$

from Year 6

14 Use a calculator: $-589 + 661$

15 Estimate, then use a calculator: -42×29

$\approx -40 \times 30 =$ _____

$=$ _____

16 Use a calculator: -46×38

17 $-63 \div (-9)$

from 1.07

18 Which keys do you press for $-6 + 4$?

Medium

Questions 19–26

19 Estimate, then use a calculator: -49×29

20 Use a calculator: $(-23)^2$

21 Which is further from 0: -13 or 9 ?

from 1.01

22 Use a calculator: $(-281 + 53) \times (-11)$

23 Use a calculator: $-1\,295 \div 37 - 296$

24 $-5 \times (-1) \times 6$

from 1.06

25 Estimate, then use a calculator: -52×22

26 Use a calculator: $(-20)^2$

Medium

Questions 27–36

- 27 Use a calculator: $(-17)^2$

- 28 Use a calculator:
 $(-121 + 50) \times (-14)$

- 29 $39 - 30 \div 6$

from Year 6

- 30 Use a calculator:
 $-1\,254 \div 38 - 282$

- 31 Start at 0. Move 2 right.
Where do you land?

from 1.02

- 32 Estimate, then use a
calculator: -45×44

- 33 Chloe pressed $(-)$ 8 $\times^2$ and got
 -64 . What keys give $(-8)^2$?
What is the answer?

- 34 $(1 + (-5)) \times (-2)$

from 1.08

- 35 Use a calculator: $(-12)^2$

- 36 A diver goes down 30 m in 3
equal stages. Write a division
for each stage.

from 1.07

Challenging

Questions 37–46

- 37 Lena pressed $(-)$ 7 $\times^2$ and got
 -49 . What keys give $(-7)^2$?
What is the answer?

- 38 Use a calculator:
 $(-330 + -12) \div (-19)$

- 39 What is $\frac{6}{8}$ of 80?

from Year 6

- 40 The temperature on Mars
changed by -15°C each hour
for 14 hours. Find the
total change.

- 41 Positive or negative? Do not
work it out: 10×5

from 1.06

- 42 Jack says -201×67 is about
 $-140\,000$. Is that a sensible
estimate? Explain.

- 43 Ali pressed $(-)$ 4 $\times^2$ and got
 -16 . What keys give $(-4)^2$?
What is the answer?

- 44 Order from largest to
smallest: $-9, 13, -26, -14$

from 1.03

- 45 Use a calculator:
 $(-346 + 458) \div (-14)$

- 46 $2 \div (7 - 8)$

from 1.08

Challenging

Questions 47–56

- 47 Use a calculator:
 $(-250 + -26) \div (-12)$

- 48 The temperature on Mars
changed by -34°C each hour
for 13 hours. Find the
total change.

- 49 $100 - 36$

from Year 6

- 50 Aisha says -293×73 is about
 $-210\,000$. Is that a sensible
estimate? Explain.

- 51 $6 + (-7)$

from 1.04

- 52 Zac pressed $(-)$ 4 $\times^2$ and got
 -16 . What keys give $(-4)^2$?
What is the answer?

- 53 Use a calculator:
 $-400 \div 16 - 288$

- 54 Positive or negative? Do not
work it out: $108 \div 9$

from 1.07

- 55 Use a calculator:
 $(-421 + 943) \div (-18)$

- 56 The temperature falls 2°C
each hour for 4 hours. Write
a multiplication for
the change.

from 1.06

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

Up, rises, gains and money in are positive. Down, falls, losses and money out are negative. Write a number sentence, then answer the question.

E1 It is -4°C . The temperature rises 9°C . What is it now?

E2 A diver at -12 m goes down 7 m , then up 15 m . Where is the diver now?

E3 The temperature falls 3°C each hour for 5 hours. What is the change?

Easy

Questions 1–7

1 It is 0°C . The temperature falls 13°C . What is it now?

$\underline{\hspace{1cm}} - \underline{\hspace{1cm}}$
 $= \underline{\hspace{1cm}}^{\circ}\text{C}$

2 Priya has $\$47$ in the bank and spends $\$68$. What is the balance?

3 A lift starts at level -4 and goes up 12 floors. Which level is it on now?

4 It is 3°C . The temperature falls 4°C . What is it now?

5 Leo has $\$77$ in the bank and spends $\$119$. What is the balance?

6 A lift starts at level -4 and goes up 5 floors. Which level is it on now?

7 It is 6°C . The temperature falls 5°C . What is it now?

Easy

Questions 8–14

8 Mia has $\$42$ in the bank and spends $\$56$. What is the balance?

$\underline{\hspace{1cm}} - \underline{\hspace{1cm}}$
 $= \underline{\hspace{1cm}}$ dollars

9 A lift starts at level -1 and goes up 8 floors. Which level is it on now?

10 It is 4°C . The temperature falls 7°C . What is it now?

11 $90 \div 9$

from Year 6

12 A diver at -5 m goes down 8 m , then up 20 m . Where is the diver now?

13 Lena has $\$34$ in the bank and spends $\$64$. What is the balance?

14 A lift starts at level -1 and goes up 11 floors. Which level is it on now?

Medium

Questions 15–20

15 A diver at -20 m goes down 7 m , then up 8 m . Where is the diver now?

16 The highest temperature was 9°C . The lowest was -11°C . What is the difference?

17 Which is further from 0: -8 or 9?

from 1.01

18 A tank's water level drops 6 cm each day for 5 days. What is the total change?

19 In three rounds of golf, Tom scored $-2, 4, 0$. What is the total score?

20 Use a calculator: $(-12)^2$

from 1.09

Medium

Questions 21–28

- 21 The highest temperature was 4°C . The lowest was -3°C . What is the difference?

- 22 A tank's water level drops 5 cm each day for 5 days. What is the total change?

- 23 $28 - 24 \div 4$

from Year 6

- 24 In three rounds of golf, Lena scored -3 , 1 , -2 . What is the total score?

- 25 Start at 6. Move 8 right. Where do you land?

from 1.02

- 26 A diver at -13 m goes down 8 m, then up 25 m. Where is the diver now?

- 27 It is 6°C at 6 pm. The temperature falls 2°C every hour. What is it 5 hours later?

- 28 $5 \times 2 - 6$

from 1.08

Challenging

Questions 29–36

- 29 It is 12°C at 6 pm. The temperature falls 4°C every hour. What is it 8 hours later?

- 30 Overnight temperatures for four nights: -1°C , -3°C , -15°C , 3°C . Find the average.

- 31 Which is largest: 6 881, 2 720, 5 329, 3 088?

from Year 6

- 32 Ivy owes \$51 and pays back \$25 a week for 4 weeks. What is the balance now?

- 33 Use the number line to find $-2 + (-3)$.



from 1.04

- 34 A plane is at 277 m. A submarine is at -110 m. How much higher is the plane?

- 35 It is 10°C at 6 pm. The temperature falls 4°C every hour. What is it 7 hours later?

- 36 Order from largest to smallest: -17 , -8 , 14 , -28

from 1.03

Challenging

Questions 37–45

- 37 Overnight temperatures for four nights: 1°C , 4°C , -8°C , -9°C . Find the average.

- 38 Ali owes \$73 and pays back \$17 a week for 3 weeks. What is the balance now?

- 39 $1\,000 - 226$

from Year 6

- 40 A plane is at 401 m. A submarine is at -229 m. How much higher is the plane?

- 41 $-4 \times (-3)$

from 1.06

- 42 It is 11°C at 6 pm. The temperature falls 2°C every hour. What is it 8 hours later?

- 43 In three rounds of golf, Lena scored -6 , 4 , 2 . What is the total score?

- 44 Which keys do you press for $-6 - 6$?

from 1.09

- 45 Overnight temperatures for four nights: -3°C , -10°C , -13°C , -10°C . Find the average.